

LLMNR, NBT-NS und mDNS-Poisoning

Introduction

The goal of this exercise was to set up a vulnerable Active Directory environment and demonstrate the risk of name-resolution poisoning within a local network. In Windows environments, when a hostname cannot be resolved via DNS (commonly with single-label names or failed DNS resolution), systems may fall back to local name-resolution protocols such as:

- **LLMNR** (Link-Local Multicast Name Resolution)
- **NBT-NS** / NBNS (NetBIOS Name Service)
- **mDNS** (Multicast DNS)

An attacker on the same network segment can spoof responses to these queries, causing the victim to authenticate to an attacker-controlled service. This can expose NetNTLMv2 (NTLMv2 challenge/response) credentials, which may be crackable offline depending on password strength and environment configuration.

Lab Environment Setup

- Domain Controller/Server → (Windows Server 2022, local IP: 10.0.0.10)
 - Domain: vulncorp.local
- Victim/Client → (Windows 10, local IP: 10.0.0.7)
 - Domain-joined user ("Anton Administrator")
- Attacker → (Kali Linux, local IP: 10.0.0.30)

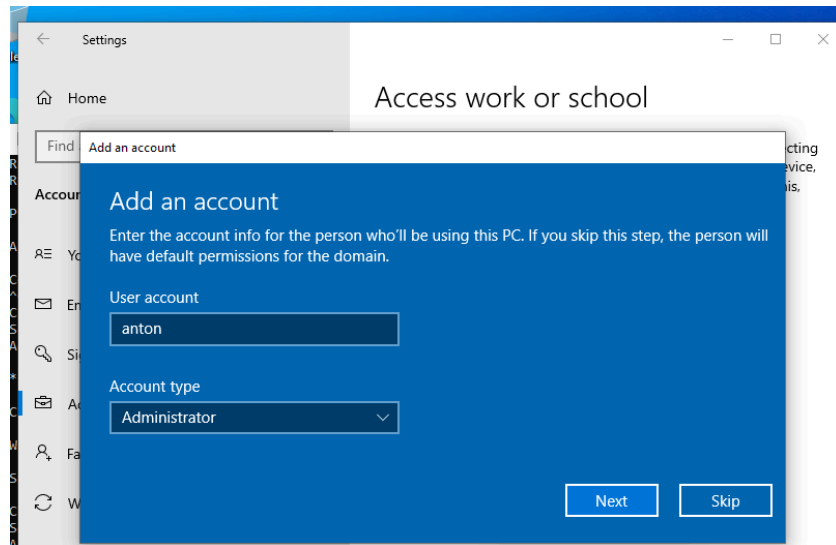
Tools

- Python3
- Responder (FOSS LLMNR, NBT-NS and MDNS poisoner)

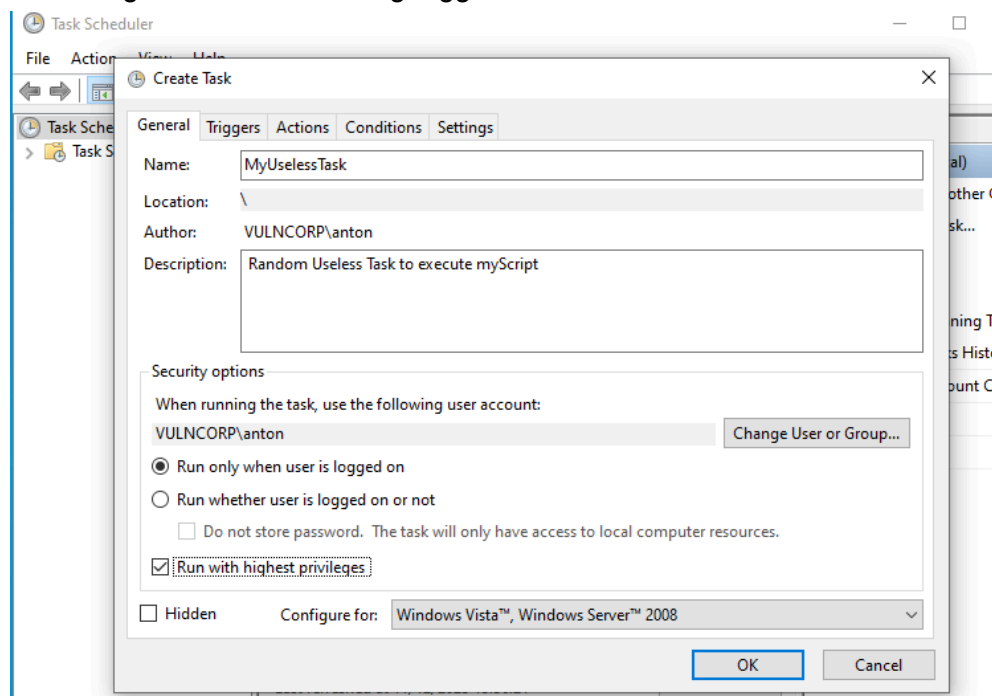
Execution

Step 1: Creating a repeatable trigger on the Windows client

To simulate a realistic recurring request for a network resource (e.g., a script or user typo), we created a scheduled task (MyUselessTask) running under the domain user VULNCORP\anton with highest privileges.



The task was configured with a recurring trigger and an action that starts a PowerShell script.

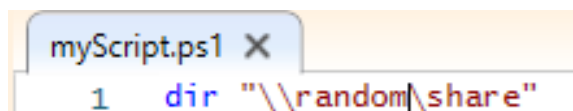


Trigger	Details	Status
Daily	At 09:00 every day - After triggered, repeat every 5 minutes for a ...	Enabled

When you create a task, you must specify the action that will occur when your task starts.

Action	Details
Start a program	C:\Users\anton.VULNCORP\Desktop\myScript.ps

The script attempts to access a non-existent UNC path `\\random\share`, which forces a name-resolution attempt.



Step 2: Verifying the name-resolution failure condition

We manually verified the trigger behavior on the client: running `dir \\random\share` results in a “path not found” error because the resource does not exist. This is the condition that causes Windows to attempt fallback name resolution when DNS cannot resolve the name.

```
PS C:\Users\anton.VULNCORP> dir \\random\share
dir : Cannot find path '\\random\share' because it does not exist.
At line:1 char:1
+ dir \\random\share
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (\\random\share:String) [Get-ChildItem], ItemNotFoundException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.GetChildItemCommand

PS C:\Users\anton.VULNCORP>
```

Step 3: Capturing authentication material via LLMNR/mDNS poisoning (lab environment)

On the attacker machine, Responder observed name-resolution fallback traffic and spoofed responses for the requested name. The client then attempted to authenticate to the attacker-controlled service, and Responder captured NetNTLMv2 (NTLMv2 challenge/response) credentials for the account VULNCORP\anton.

[illegible]

Note: This is not the stored NTLM password hash; it is the challenge/response material produced during NTLM authentication.

To illustrate the risk of weak passwords, the captured NetNTLMv2 challenge/response was tested offline against a small lab wordlist. A match was found for the lab account, demonstrating that captured NetNTLMv2 material may be crackable when weak passwords are used.

[illegible]

```
(kali㉿kali)-[~/Responder]
└─$ john --wordlist=wordlist.txt hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (netntlmv2, NTLMv2 C/R [MD4 HMAC-MD5 32/64])
Will run 2 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
anton2 (anton)
1g 0:00:00.00 DONE (2025-12-11 12:10) 25.00g/s 175.0p/s 175.0c/s 175.0C/s franz..sebastian
Use the "--show --format=netntlmv2" options to display all of the cracked passwords reliably
Session completed.
```

Mitigation

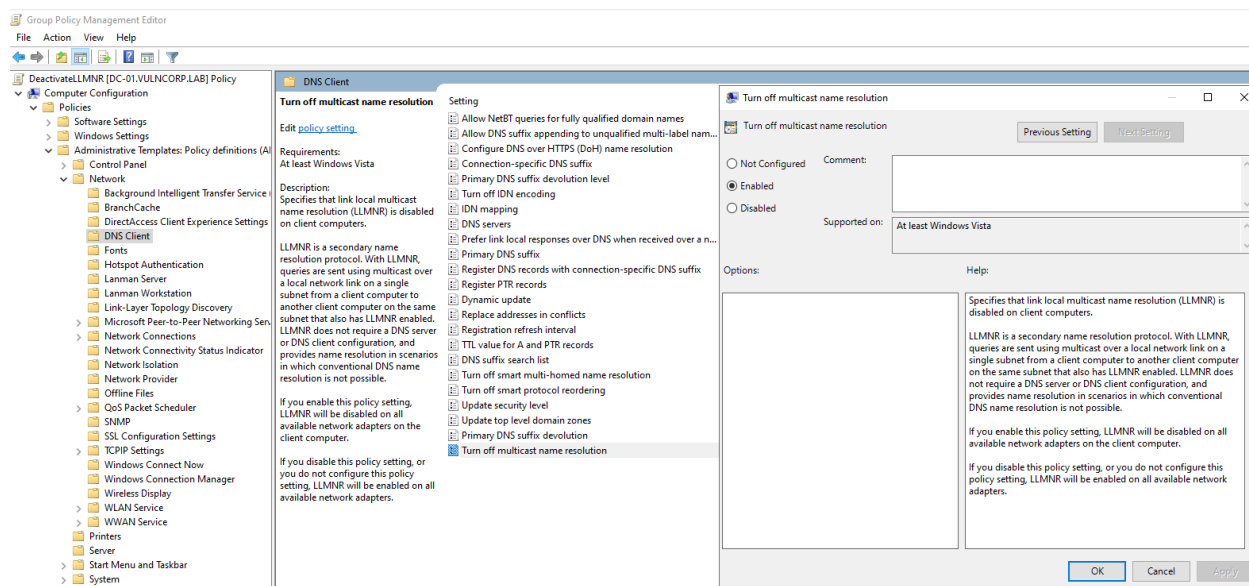
- **Disable LLMNR via GPO (Domain-Wide)**

- Computer Configuration → Administrative Templates → Network → DNS Client
- Setting: **Turn off multicast name resolution** → Set to **Enabled**

- **Additional Hardening**

Because the lab topic includes NBT-NS/NBNS and mDNS, disabling LLMNR alone may not remove all poisoning opportunities. Additional hardening may include:

- Reducing or disabling NetBIOS over TCP/IP where not required.
- Disabling mDNS where not needed (platform/config dependent).
- Hardening authentication to reduce impact (e.g., limiting NTLM where possible, enforcing SMB signing, and applying relay mitigations).



References

<https://github.com/lgandx/Responder>

<https://github.com/FelixSchuster/VulnAD>